

# Project 1: Problem Statement & Initial Concept

*Working Project Title: Kept — A Calm Commitment App*

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## Problem Statement

Conventional to-do and habit apps fail the everyday promise-keeper in three compounding ways. First, they induce overwhelm: they invite users to externalize an unbounded list of obligations, and a large share of those items are simply never finished. Second, long lists collide with hard cognitive limits: working memory holds only about four chunks (Cowan, 2001), far fewer than Miller’s famous “seven, plus or minus two,” and unfulfilled goals impose a persistent mental tax, intruding on unrelated tasks (Masicampo & Baumeister, 2011). Third, these tools are not designed to foster social reinforcement: they treat follow-through as solitary bookkeeping, ignoring strong evidence that being witnessed raises commitment, while the apps that do add social pressure tend to reach for punitive stakes and streak-shame, which can crowd out the intrinsic motivation that makes a personal promise meaningful in the first place (Deci et al., 1999). The result is a category where a median of 70% of users abandon within 100 days (Lawes-Wickwar et al., 2024). This problem is personal: I have cycled through these tools myself, and the gap between what matters to me and what survives a busy week is the gap this project addresses.

## “How Might We...”

*Some people struggle to stick to what matters to them when life gets busy and no one’s checking. How might we make it easier for someone to keep everyday promises to themselves, in the context of a normal day?*

## Scope: What John is (trying to) Build

A desktop + mobile app built around one constraint and one loop. The constraint: **One active promise at a time** no lists, no dashboard (grounded in

Cowan’s  $4 \pm 1$  working-memory limit and the “Hrair limit” metaphor from McConnell’s *Rapid Development* (McConnell, 1996)). The loop: Make a promise phrased as an implementation intention (“when X, I will Y” (Gollwitzer & Sheeran, 2006))  $\rightarrow$  Act  $\rightarrow$  Mark It  $\rightarrow$  A Brief Closing Ritual. Three design commitments distinguish it from the category:

- **The Living Artifact.** Instead of streaks or scores, follow-through feeds a generative visual/ambient piece (p5.js / algorithmic music) that flourishes or quietly wilts - calm, peripheral feedback in the tradition of Weiser & Brown’s calm technology (Weiser & Brown, 1996), not a guilt meter.
- **Witness mode.** Social presence is a property of a single promise, invited never forced: one trusted person sees one silent dot — kept or not — and cannot comment. Being seen, without being supervised.
- **Break = Reflect, Not Punishment.** A missed promise opens a 10-second “what got in the way?” inquiry instead of a broken streak.

Prototype plan for Weeks 6-7: A working desktop app demonstrating the full Solo loop and Living Artifact, paired with an Android emulator acting as the Witness — so the class can watch a promise get marked on the desktop and the witness’s dot change on the phone in real time. At least one informal user feedback session before the Week 6 crit, focused on whether the single-promise constraint feels liberating or limiting.

## Scope: What’s Intentionally Left Out

- **Later social modes.** The full vision includes Reciprocal (mutual pacts), Anon (anonymous solidarity), and Circle (small-group weekly ritual) modes; all are out of scope for this project. Solo + Witness is enough to prove the thesis that one core loop can carry graduated social presence.
- **The habit-app arsenal.** No task lists, streak counters, leaderboards, monetary stakes, gamified rewards, or re-engagement notifications — these are the documented churn and motivation-crowding mechanisms the design is built against.
- **Production infrastructure.** No accounts, app-store deployment, or scaled backend; the desktop-emulator pairing will use a lightweight local sync sufficient to demonstrate the interaction.

## References

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